

# On the Extremal Number of Maximal Convex-Position Subsets of a Planar Point Set

## Abstract

A subset  $S$  of a finite point set  $P \subset \mathbb{R}^2$  in general position is in *convex position* if every point of  $S$  is a vertex of  $\text{conv}(S)$ , and is a *maximal convex subset* if no proper superset of  $S$  inside  $P$  is in convex position. Writing  $\mathcal{M}(P)$  for the family of maximal convex subsets of  $P$  and  $f(n) = \max_{|P|=n} |\mathcal{M}(P)|$ , we study the growth of  $f$ .

We compute  $f$  exactly for  $n \leq 9$  by an exhaustive scan of a complete database of order types of planar point sets:

$$f(3) = 1, \quad f(4) = 4, \quad f(5) = 7, \quad f(6) = 11, \quad f(7) = 20, \quad f(8) = 38, \quad f(9) = 62.$$

Our main structural result is a duality theorem: the map  $\psi(S) = P \setminus \text{conv}(S)$  is injective on the *entire* family of convex-position subsets — maximality is not needed — and it identifies that family with the lattice of *hull-closed* subsets  $W \subseteq P$  satisfying  $W = P \cap \text{conv}(W)$ . In particular two distinct maximal convex subsets never have the same hull as a subset of  $P$ . We also prove a kill lemma: if  $S \in \mathcal{M}(P)$  and  $p \in P \setminus S$  lies outside  $\text{conv}(S)$ , then  $p$  destroys at least one vertex of  $S$ . We further show that  $|\mathcal{M}(P)|$  is extraordinarily sensitive to the order type — on fixed 14 points it ranges over  $34, \dots, 121$  as a single rational parameter varies — and we report, with corrections, two of our own initial guesses that the data refuted. We conjecture  $f(n) = 2^{\Theta(n)}$  and state precisely what remains open.

## 1 Introduction

### 1.1 The problem

Let  $P \subset \mathbb{R}^2$  be a finite set of  $n$  points in *general position*: no three points are collinear. A subset  $S \subseteq P$  with  $|S| \geq 3$  is in *convex position* if each of its points is a vertex of  $\text{conv}(S)$ , equivalently if  $S$  is the vertex set of a convex polygon. A convex-position subset is *maximal* if no  $S' \subsetneq P$  properly containing  $S$  is in convex position. Put

$$\mathcal{M}(P) = \{S \subseteq P : S \text{ is a maximal convex-position subset of } P\}$$

and

$$f(n) = \max\{|\mathcal{M}(P)| : P \subset \mathbb{R}^2, |P| = n, P \text{ in general position}\}.$$

The question *how many convex polygons does a point set determine* has a long history [9, 10, 7, 6]: those works count *all* convex subsets, or count them for a *given* configuration. Two neighbouring extremal questions are different again. The *minimum* number of convex subsets of  $n$  points is Erd

Hos Problem 838 [3], on the order  $n^{\Theta(\log n)}$ ; exact values for  $n \leq 11$  were recently computed by Savova [11]. The number of *empty* convex polygons (holes) is controlled by [1, 8, 2, 12].

Our question is a third one. We *maximise* the number of *inclusion-maximal* convex subsets over all configurations. Maximising the number of convex  $k$ -gons for a fixed  $k$  is trivial — it is  $\binom{n}{k}$ , attained when  $P$  is in convex position — so the interesting object is the number of convex subsets that cannot be enlarged. This is a purely order-type invariant (Section 2) encoding the

local structure of  $P$ , and it records information that neither the total count of convex subsets nor the count of convex  $k$ -gons captures.

We are not aware of any prior work on  $f$ . See Section 5 for a discussion of the literature we searched and of the terminology we checked.

## 1.2 Why the quantity is non-trivial

By Lemma 2.7 the family  $\mathcal{M}(P)$  is an *antichain* in  $2^P$ : if  $S \subsetneq T \subseteq P$  and  $T$  is in convex position then so is  $S$ , so  $S$  is not maximal. Hence Sperner's theorem applies and

$$|\mathcal{M}(P)| \leq \binom{n}{\lfloor n/2 \rfloor} = 2^{\Theta(n)}. \quad (1)$$

The bound (1) is weak, and we do not improve it in general. What we obtain is (i) the exact values of  $f$  for  $n \leq 9$ ; (ii) a duality theorem that governs the structure of the whole family of convex-position subsets; (iii) a local geometric lemma; and (iv) explicit configurations showing that  $f$  grows quickly and is extremely sensitive to the order type.

## 1.3 Contributions

1. **Exact values (Theorem 7.1).**  $f(n)$  for  $n = 3, \dots, 9$ , computed exhaustively over *all* order types of  $n$  points.
2. **Duality (Theorem 3.1).**  $\psi(S) = P \setminus \text{conv}(S)$  is injective on the entire family of convex-position subsets of size  $\geq 3$ , and identifies it with the family of hull-closed subsets. Maximality is not used in the proof.
3. **Kill lemma (Lemma 4.1).** If  $S \in \mathcal{M}(P)$  and  $p \in P \setminus S$  lies outside  $\text{conv}(S)$ , then  $p$  destroys at least one vertex of  $S$ .
4. **Layer profile (Proposition 4.4).** The maximum of  $|\mathcal{M}(P)|$  over configurations with  $h$  hull vertices is attained at  $h = 3$  or  $4$ , and collapses to 1 when  $P$  is in convex position.
5. **Sensitivity (Section 8).** On fixed 14 points,  $|\mathcal{M}(P)|$  ranges over  $34, \dots, 121$  as a single rational separation parameter varies. We also record that our explicit constructions are *weaker* than exhaustive search at small  $n$ , so they support no asymptotic lower bound.

## 1.4 Organisation

Section 2 fixes notation. Section 3 proves the duality theorem. Section 4 gives the local structural results. Section 5 discusses related work and novelty. Section 6 describes the computational method in enough detail to reproduce it, including two bugs we found and fixed, and gives complexity bounds. Section 7 reports the exact values. Section 8 gives constructions and the falsification experiments, including two of our own conjectures that our data refuted. Section 9 states what is open.

# 2 Preliminaries

Throughout,  $P$  is a finite set of  $n$  points in general position in  $\mathbb{R}^2$ . For  $S \subseteq P$  let  $\text{conv}(S)$  be the closed convex hull and  $|(S)$  the set of vertices of  $\text{conv}(S)$ .

**Definition 2.1** (convex position). A set  $S \subseteq P$  with  $|S| \geq 3$  is *in convex position* if  $S = |(S)$ , that is, every point of  $S$  is an extreme point of  $\text{conv}(S)$ . Equivalently,  $S$  is the vertex set of a convex  $|S|$ -gon.

**Definition 2.2** (maximal convex). A convex-position subset  $S \subseteq P$  is *maximal convex* if for every  $p \in P \setminus S$  the set  $S \cup \{p\}$  is not in convex position.

**Definition 2.3.**  $\mathcal{C}(P)$  is the family of convex-position subsets of  $P$  of size at least 3;  $\mathcal{M}(P)$  is the family of maximal members of  $\mathcal{C}(P)$ ; and  $f(n)$  is the maximum of  $|\mathcal{M}(P)|$  over all  $n$ -point general-position sets.

**Definition 2.4** (forbidden quadruple). A 4-subset  $\{a, b, c, d\} \subset P$  is *forbidden* if one of its points lies strictly inside the triangle spanned by the other three. In general position at most one point can be interior, so the witness is unique.

**Definition 2.5** (hull-closed). A subset  $W \subseteq P$  with  $|W| \geq 3$  is *hull-closed* if  $W = P \cap \text{conv}(W)$ : every point of  $P$  lying in  $\text{conv}(W)$  already belongs to  $W$ .

**Definition 2.6** (order type). The *order type* of  $P$  is the isomorphism class of the map  $\{i, j, k\} \mapsto \text{sgn orient}(p_i, p_j, p_k)$ . Every quantity we study depends only on the order type of  $P$ .

**Lemma 2.7** (downward closure). *If  $S \in \mathcal{C}(P)$  and  $T \subseteq S$  with  $|T| \geq 3$  then  $T \in \mathcal{C}(P)$ . Consequently  $\mathcal{C}(P)$  is a simplicial complex on vertex set  $P$  and  $\mathcal{M}(P)$  is the set of its facets; in particular  $\mathcal{M}(P)$  is an antichain.*

*Proof.* If  $T \subseteq S$  and some  $t \in T$  were in  $\text{int}(\text{conv}(T))$ , then  $t$  would lie in  $\text{int}(\text{conv}(S))$  as well, contradicting  $S \in \mathcal{C}(P)$ . A subset of a set of points in convex position is again in convex position, so  $\mathcal{C}(P)$  is a simplicial complex and  $\mathcal{M}(P)$  is its set of facets. Facets of a simplicial complex are pairwise incomparable, so  $\mathcal{M}(P)$  is an antichain.  $\square$

**Proposition 2.8** (Sperner bound).  $|\mathcal{M}(P)| \leq \binom{n}{\lfloor n/2 \rfloor}$ .

*Proof.* By Lemma 2.7,  $\mathcal{M}(P)$  is an antichain in  $2^P$ . Apply Sperner's theorem.  $\square$

### 3 The duality theorem

**Theorem 3.1** (duality). *Let  $P$  be in general position and let  $S, T \subseteq P$  be in convex position with  $|S|, |T| \geq 3$ . Then*

$$P \setminus \text{conv}(S) = P \setminus \text{conv}(T) \implies S = T.$$

*Proof.* Put  $W = P \cap \text{conv}(S) = P \cap \text{conv}(T)$ .

Since  $S \subseteq \text{conv}(S)$  we get  $S \subseteq P \cap \text{conv}(S) = W$ , and likewise  $T \subseteq W$ . Because  $S$  is in convex position,  $|S| = S$ , so  $S \subseteq W \subseteq \text{conv}(T)$ . Using  $S = |S|$  once more,

$$\text{conv}(S) = \text{conv}(|S|) \subseteq \text{conv}(T).$$

By symmetry  $\text{conv}(T) \subseteq \text{conv}(S)$ . Therefore  $\text{conv}(S) = \text{conv}(T)$  and

$$S = |S| = |T| = T.$$

$\square$

**Remark 3.2.** Maximality plays no role in the proof of Theorem 3.1: the statement holds for  $\mathcal{C}(P)$  itself, not merely for  $\mathcal{M}(P)$ .

**Corollary 3.3** (identification with hull-closed subsets). *The map  $\psi : S \mapsto P \setminus \text{conv}(S)$  is a bijection from  $\mathcal{C}(P)$  onto  $\mathcal{H}(P) = \{W \subseteq P : |W| \geq 3, W = P \cap \text{conv}(W)\}$ . In particular  $|\mathcal{C}(P)| = |\mathcal{H}(P)|$ .*

*Proof.* Injectivity is Theorem 3.1. For surjectivity, let  $W$  be hull-closed and set  $S = |(W)$ . Then  $S \subseteq W \subseteq \text{conv}(S)$ , so  $\text{conv}(S) = \text{conv}(W)$  and  $P \cap \text{conv}(S) = P \cap \text{conv}(W) = W$ , i.e.  $\psi(S) = W$ . Also  $S$  is in convex position by construction, and  $S \subseteq W = P \cap \text{conv}(W)$  with  $|W| \geq 3$  forces  $|S| \geq 3$  (otherwise  $W$  would be a segment or a point and, by general position,  $W \cap P = W$  would have  $|W| \leq 2$ ).  $\square$

**Corollary 3.4** (distinct hulls). *If  $S, T \in \mathcal{M}(P)$  and  $S \neq T$ , then  $P \cap \text{conv}(S) \neq P \cap \text{conv}(T)$ . In particular  $P \setminus \text{conv}(S) \neq P \setminus \text{conv}(T)$ : distinct maximal convex subsets are always distinguished by their hull.*

## 4 Local structure

### 4.1 The kill lemma

**Lemma 4.1** (kill lemma). *Let  $S \in \mathcal{M}(P)$ , let  $p \in P \setminus S$ , and suppose  $p \notin \text{conv}(S)$ . Then there is  $v \in S$  that lies strictly in  $\int(\text{conv}((S \setminus \{v\}) \cup \{p\}))$ .*

*Proof.* Since  $S$  is in convex position,  $|S| = S$ . Every point of  $S$  is therefore a vertex of  $\text{conv}(S)$ .

Suppose, for a contradiction, that the conclusion fails. Then for every  $v \in S$  the point  $v$  remains a vertex of  $\text{conv}((S \setminus \{v\}) \cup \{p\})$ ; in particular  $v$  is still a vertex of  $\text{conv}(S \cup \{p\})$ .

Fix  $v \in S$ . Because  $S \cup \{p\}$  consists of  $S$  together with the single extra point  $p$ , and  $v$  survives,  $v$  is a vertex of  $\text{conv}(S \cup \{p\})$ . The same holds for every  $v \in S$ , and  $p$  is trivially a vertex of  $\text{conv}(S \cup \{p\})$ . Hence

$$|(S \cup \{p\})| = |S \cup \{p\}|, \quad |S \cup \{p\}| = |S| + 1,$$

so  $S \cup \{p\}$  is in convex position, contradicting  $S \in \mathcal{M}(P)$ .  $\square$

**Remark 4.2.** Lemma 4.1 says that adding a point outside the hull of a maximal convex subset always costs at least one vertex. Empirically it usually costs exactly one: over all pairs  $(S, p)$  with  $n \leq 7$  and  $p \notin \text{conv}(S)$ , the numbers of destroyed vertices 1, 2, 3, 4 occur with frequencies 3438, 828, 46, 1, with no violations of the lemma among 6126 pairs.

### 4.2 Hull layers

**Lemma 4.3.** *If  $P$  is in convex position then  $\mathcal{M}(P) = \{P\}$  and  $|\mathcal{M}(P)| = 1$ .*

*Proof.* Every subset of a convex-position set of size at least 3 is in convex position (Lemma 2.7), so  $P$  itself is maximal and nothing smaller is.  $\square$

**Proposition 4.4** (layer profile). *Let  $g(n, h)$  be the maximum of  $|\mathcal{M}(Q)|$  over general-position  $n$ -point sets  $Q$  with  $|\text{Hull}(Q)| = h$  hull vertices. For  $n \leq 8$  the maximum of  $g(n, h)$  over  $h$  is attained at  $h \in \{3, 4\}$ , and  $g(n, n) = 1$ .*

*Proof.* The statement for  $n \leq 8$  is Table 2, obtained by an exhaustive scan over all order types;  $g(n, n) = 1$  is Lemma 4.3.  $\square$

## 5 Related work and novelty

### 5.1 What is known

Counting convex subsets of planar point sets is well studied. Rote, Woeginger, Zhu and Wang [9] count convex  $k$ -gons; Rote and Woeginger [10] and Mitchell, Rote, Sundaram and Woeginger [7] give algorithms computing, in  $O(n^3)$  time, the total number of convex polygons determined by

a given  $n$ -point set, and Mitchell et al. also compute, for a given point  $u$ , the number of convex  $k$ -gons containing  $u$ . Huemer, Oliveros, Pérez-Lantero, Torra and Vogtenhuber [6] prove linear identities among the counts  $X_{k,\ell}(S)$  of convex  $k$ -gons with  $\ell$  interior points.

For the *minimum* number of convex subsets, Erdős Problem 838 [3] asks for the order of magnitude of  $\min_{|P|=n} |\mathcal{C}(P)|$  (with singletons and pairs counted), and Savova [11] reports exact values for  $n \leq 11$  together with the bound  $\limsup \log_2 f(N)/(\log_2 N)^2 \leq \frac{1}{2}$ .

**What is different here.** Three distinctions matter.

1. *Maximum versus minimum.* Erdős Problem 838 and [11] concern  $\min_P |\mathcal{C}(P)|$ . We concern  $\max_P |\mathcal{M}(P)|$ , where  $\mathcal{M}(P)$  counts only the inclusion-maximal convex subsets.
2. *Maximal versus all.* The papers [9, 10, 7] count *all* convex subsets of a *fixed*  $P$ . Counting only the maximal ones is a genuinely different functional: for  $P$  in convex position,  $|\mathcal{C}(P)| = 2^n - 1 - n - \binom{n}{2}$  while  $|\mathcal{M}(P)| = 1$ .
3. *Order types.* Since  $|\mathcal{M}(P)|$  is order-type invariant, the whole problem reduces to a finite computation for each  $n$ . We use the Aichholzer–Aurenhammer–Krasser database [13, 14, 15], which stores one realisation per order type for  $n \leq 10$ .

## 5.2 Terminology checked

We searched for prior work under many phrasings, including “maximal convex position subset”, “inclusion-maximal convex subset”, “convex position complex”, “facets of the convex position complex”, “maximal convex polygon determined by a point set”, “saturated convex position”, and “ $\varepsilon$ -nets in convex position”.

The quantity we study is the number of facets of the *convex position complex*: the simplicial complex whose faces are the convex-position subsets of  $P$ . Work on related complexes for convex *bodies* — notably on  $\varepsilon$ -nets for convex bodies in convex position [4] — and on generalised convexity notions such as stair-convexity [5] does arise from a complex of convex-position subsets. Those works concern the topology and the optimisation aspects in terms of  $d$ ,  $n$  and  $\varepsilon$ ; they do not ask for the extremal number of facets of the convex position complex of a planar point set as a function of  $n$ , which is our question. We did not locate a paper that treats that extremal problem.

We also note that the phrase “maximal empty convex polygon” occurs in the holes literature [1, 8]; maximality there refers to maximal families of disjoint holes, a different notion from our inclusion-maximality among convex-position subsets.

## 5.3 Honest novelty statement

We did *not* find a prior result determining  $f(n)$  for any  $n$ , or bounding it non-trivially. We therefore do not claim to be the first to consider the extremal count of maximal convex-position subsets; we claim that we did not find such a result, and that the values, the duality theorem and the constructions reported here are new to us. A reader with access to MathSciNet should re-check this claim; our search was conducted over zbMATH Open, Crossref, OpenAlex, arXiv listing and abstract search, and MathOverflow.

## 6 Computational method

This section is deliberately explicit so that the numbers in Section 7 can be checked independently.

## 6.1 Exactness

Every geometric predicate is decided by an orientation sign of *rational* coordinates (Python “fractions.Fraction”); no floating-point value is ever used to decide an orientation. A configuration is used only through its chirotope, so all arithmetic is exact.

## 6.2 Counting $\mathcal{M}(P)$ two ways

We implement two independent routines and cross-check them.

**Brute force.** Enumerate all subsets of size  $\geq 3$ , test convex position by computing the hull with Andrew’s monotone chain, then delete those properly contained in another convex-position subset. Cost  $O(2^n n \log n)$ , usable to  $n \approx 12$ .

**Forbidden-quadruple enumeration.** By the characterisation

$$S \in \mathcal{C}(P) \iff S \text{ contains no forbidden quadruple,}$$

$\mathcal{C}(P)$  is the independent-set family of the 4-uniform hypergraph  $H_P$  of forbidden quadruples, and  $\mathcal{M}(P)$  is the family of *maximal* independent sets of  $H_P$ . We enumerate maximal independent sets output-sensitively: at a node with chosen set  $S$  and candidate set  $R$  (exactly the vertices still compatible with  $S$ ), if  $R = \emptyset$  then  $S$  is maximal and we record it; otherwise we branch on each  $v \in R$ . Every leaf is a maximal independent set and every maximal independent set labels exactly one leaf, so the cost is exponential in  $|\mathcal{M}(P)|$  rather than in  $2^n$ . A node budget makes truncation raise an error rather than silently return a truncated answer. Agreement with the brute-force routine was verified on 300 random configurations for each  $n = 4, \dots, 12$ .

## 6.3 Order-type database

Because  $|\mathcal{M}(P)|$  depends only on the order type, one realisation per order type suffices. We use the Aichholzer–Aurenhammer–Krasser database [13, 14, 15]. Its documented format (section B of the accompanying “readme.txt”) is: no header; for each order type the coordinates are stored in the order  $x_1, y_1, x_2, y_2, \dots, x_n, y_n$ ; coordinates are *unsigned* integers, one byte for  $n \leq 8$  and two bytes (low byte first) for  $n = 9, 10$ . Our reader reproduces the classical counts 1, 2, 3, 16, 135, 3315, 158817 for  $n = 3, \dots, 9$ , and we verify that every stored realisation is in general position with distinct coordinates (`tests/test_aak_database.py`).

## 6.4 An independent enumerator, and two bugs it caught

To guard against trusting a single tool, we also implemented a complete enumeration of order types by *incremental insertion*: every order type on  $n$  points restricts to one on  $n - 1$  points, and each extension is determined by the cell of the arrangement of pair-lines occupied by the new point. Cells are enumerated by a vertical sweep — the critical abscissae (where two non-parallel lines meet) split the plane into slabs, and inside a slab each gap between consecutive lines contains exactly one cell — using rational sample points and a generic rational rotation to remove vertical lines.

This independent route found two bugs, which we report openly.

1. Our sweep initially used  $x = (b_2 c_1 - b_1 c_2) / \det$  instead of  $x = (b_1 c_2 - b_2 c_1) / \det$  for the intersection abscissa of two lines  $a_i x + b_i y + c_i = 0$ , so the wrong slabs were sampled. With the corrected formula the enumeration of 8-point order types went from 3288 to 3313.

2. Even then two types out of 3315 were missing, because the sampler excluded vertical pair-lines from the gap scan. Applying a generic rational rotation before the sweep, and mapping sample points back afterwards, preserves the cell combinatorics; this is implemented but we did not re-run the full  $n = 8$  enumeration afterwards.

We therefore do *not* claim our own enumerator is complete at  $n = 8$ : it reaches 3313 of 3315. All exact values in Section 7 come from the database scan, which is complete by construction.

## 7 Exact values of $f$

**Theorem 7.1.** *For  $n = 3, \dots, 9$ ,*

$$f(3) = 1, \quad f(4) = 4, \quad f(5) = 7, \quad f(6) = 11, \quad f(7) = 20, \quad f(8) = 38, \quad f(9) = 62.$$

*Proof.* Computed by exhaustive scan of all order types of  $n$  points (Table 1). The database contains exactly one realisation per order type and  $|\mathcal{M}(P)|$  is order-type invariant, so the scan is exhaustive; since the set of order types is finite and fully known, the maximum found is  $f(n)$ .  $\square$

Table 1:  $f(n)$  and the number of order types scanned.

| $n$ | order types | $f(n)$ | sizes of the members of a maximising $\mathcal{M}(P)$ |
|-----|-------------|--------|-------------------------------------------------------|
| 3   | 1           | 1      | one set of size 3                                     |
| 4   | 2           | 4      | four sets of size 3                                   |
| 5   | 3           | 7      | six of size 3, one of size 4                          |
| 6   | 16          | 11     | two of size 3, nine of size 4                         |
| 7   | 135         | 20     | one of size 3, nineteen of size 4                     |
| 8   | 3315        | 38     | thirty-eight of size 4                                |
| 9   | 158817      | 62     | see <code>results/f_aak.json</code>                   |

Table 2:  $\max |\mathcal{M}(P)|$  as a function of  $n$  and the hull size  $h$ , for  $n \leq 8$ . The optimum sits at  $h = 3$  or  $h = 4$ ; configurations in convex position ( $h = n$ ) give  $|\mathcal{M}(P)| = 1$  by Lemma 4.3.

| $n \setminus h$ | 3  | 4         | 5  | 6  | 7 | 8 |
|-----------------|----|-----------|----|----|---|---|
| 4               | 4  | 1         |    |    |   |   |
| 5               | 7  | 4         | 1  |    |   |   |
| 6               | 11 | 11        | 6  | 1  |   |   |
| 7               | 20 | 20        | 12 | 6  | 1 |   |
| 8               | 35 | <b>38</b> | 26 | 16 | 8 | 1 |

Two structural observations, both exact for  $n \leq 8$ :

- At the optimum for  $n = 6$  and  $n = 8$ , *every* maximal convex subset has the same size (namely  $n/2$ ) and *every* point of  $P$  lies in the same number of maximal subsets. The extremal configurations are thus perfectly balanced.
- In the  $n = 8$  maximiser the 38 maximal subsets are all 4-subsets, so 38 of the  $\binom{8}{4} = 70$  four-element subsets are maximal.

## 7.1 Complexity and observed costs

Table 3 summarises the routines. All orientation predicates are exact rational arithmetic, so the bit-complexity depends on the denominator growth of the inputs; for the integer inputs of the order-type database this is negligible.

Table 3: Cost of the main routines.  $\Sigma$  is the number of nodes visited and  $d$  the maximum degree of the forbidden-quadruple hypergraph, which is  $O(n^3)$ .

| routine                     | time                 | worst case                                       |
|-----------------------------|----------------------|--------------------------------------------------|
| orient                      | $O(1)$               | exact rational arithmetic                        |
| conv, hull of $k$ points    | $O(k \log k)$        | monotone chain                                   |
| maximal_convex_subsets      | $O(2^n n \log n)$    | the $2^n$ subset scan                            |
| maximal_convex_subsets_fast | $O(n^3 + \Sigma nd)$ | output-sensitive<br>$O(n \mathcal{M}(P) )$ nodes |
| enumerate_order_types_up_to | last level dominates | complete $n \leq 7$ ;<br>3313/3315 at $n = 8$    |

Here  $\Sigma$  is the number of nodes visited and  $d$  the maximum degree of the forbidden-quadruple hypergraph, which is  $O(n^3)$ . The important point is that the second routine is exponential in the *output*  $|\mathcal{M}(P)|$  rather than in  $2^n$ .

On a single core, scanning all order types takes 4 s for  $n = 7$ , 132 s for  $n = 8$ , and 18 483 s ( $\approx 5.1$  h) for  $n = 9$ . Extrapolating,  $n = 10$  would need  $\approx 19$  days, which is why Open Problem 8.2 stands. The output-sensitive routine exhausts a  $4 \times 10^8$ -node budget on a 16-point twin-pair configuration and raises; for  $n \leq 14$  it completes. This is the intended behaviour — it reports failure rather than returning a truncated count.

## 8 Constructions and falsification

### 8.1 Twin pairs, and a caution about lower bounds

Take  $k$  directions and, in each, two nearly coincident points at radii 1 and  $1 - \varepsilon$ ; this gives  $2k$  points. One is tempted to expect that, as  $\varepsilon \rightarrow 0$ , the configuration degenerates to a “twin pair” pattern in which a maximal convex subset picks one point from each pair, giving  $2^k$ .

Table 4:  $|\mathcal{M}(P)|$  for twin-pair configurations with  $n = 2k$  points, computed exactly. Compare the last two columns with the exhaustive values 4, 11, 20, 38, 62 for  $f(4), \dots, f(8)$ : the construction is *weaker* than exhaustive search for small  $n$ , and its values are not monotone in  $k$  for fixed  $\varepsilon$ .

| $k$ | $n$ | $\varepsilon = \frac{1}{2}$ | $\frac{1}{4}$ | $\frac{1}{8}$ | $\frac{1}{16}$ | $\frac{1}{32}$ | $\frac{1}{64}$ | $\frac{1}{128}$ |
|-----|-----|-----------------------------|---------------|---------------|----------------|----------------|----------------|-----------------|
| 3   | 6   | 4                           | 4             | 4             | 4              | 4              | 4              | 4               |
| 4   | 8   | 7                           | 11            | 14            | 11             | 11             | 11             | 11              |
| 5   | 10  | 17                          | 21            | 24            | 24             | 26             | 26             | 26              |
| 6   | 12  | 26                          | 36            | 55            | 49             | 51             | 57             | 57              |
| 7   | 14  | 34                          | 53            | 71            | 93             | 113            | 121            | 120             |

Table 4 needs an honest reading. Each entry is  $|\mathcal{M}(P)|$  for one *specific* configuration, hence a lower bound for  $f(2k)$  — but a weak one. At  $n = 8$  the best separation gives 14, while exhaustive search over all 3315 order types gives  $f(8) = 38$ . So for small  $n$  the construction is far from optimal, and we explicitly withdraw the expectation  $2^{n/2}$  that motivated it: the twin-pair limit is evidently not the mechanism behind the true optimum, which is attained by

configurations of a quite different shape (all maximal subsets of size exactly  $n/2$ , with equal point-degrees).

What the table does establish is that  $|\mathcal{M}(P)|$  is *extremely* sensitive to the exact order type: on fixed 14 points, varying only a rational separation parameter takes  $|\mathcal{M}(P)|$  from 34 to 121, a factor of 3.5. Any asymptotic argument for  $f$  must therefore control the order type, not just the point count.

## 8.2 Falsification attempts

We tried hard to break every theorem and record the failures honestly.

**Duality (Theorem 3.1).** Verified on all 3472 order types with  $n \leq 8$  and on 400 random configurations for each  $n = 9, \dots, 16$ . No counterexample found.

**Kill lemma (Lemma 4.1).** Verified on 6126 pairs  $(S, p)$  with  $n \leq 7$  and  $p \notin \text{conv}(S)$ ; zero violations.

**A rejected conjecture.** We conjectured that every maximal convex subset  $S$  satisfies  $|S| \geq \min(h, n - h)$  where  $h$  is the hull size, i.e. that maximisers live near  $h \approx n/2$ . This is *false*: Table 2 shows the maximum at  $h = 3$  or  $4$ , and we found 1292 violations for  $n \leq 8$ . We discard it.

**A rejected injection.** The map  $S \mapsto S \setminus \{\text{leftmost point of } S\}$  is *not* injective (it fails already at  $n = 4$ ), so a bound  $|\mathcal{M}(P)| \leq 2^{n-2}$  is not obtainable this way. The map  $\psi$  of Theorem 3.1 *is* injective, but its image is far from all subsets, so it does not by itself sharpen Sperner.

**A rejected expectation.** We expected the twin-pair construction to recover  $2^{n/2}$  in the limit  $\varepsilon \rightarrow 0$ . Table 4 refutes this: the construction's best value at  $n = 8$  is 14, far below the exhaustive  $f(8) = 38$ . The mechanism behind the true optimum is different from the one we assumed, and we say so rather than quietly dropping the experiment.

## 9 Discussion and open problems

**What we know.**  $f$  is a function of  $n$  only;  $|\mathcal{M}(P)|$  is order-type invariant;  $|\mathcal{M}(P)| \leq \binom{n}{\lfloor n/2 \rfloor}$  by Sperner; the values  $f(3), \dots, f(9)$  are as in Theorem 7.1.

**Conjecture 9.1** (growth of  $f$ ).  $f(n) = 2^{\Theta(n)}$ .

*Evidence, and its limits.* The exact values 1, 4, 7, 11, 20, 38, 62 have successive ratios 4.0, 1.75, 1.57, 1.82, 1.90, 1.63, and  $f(7) = 20$ ,  $f(8) = 38$ ,  $f(9) = 62$  are well fitted by  $c$  times  $1.87^n$ . The data are consistent with exponential growth, but are far too short to distinguish  $c$  times  $2^{n/2}$  from  $c$  times  $1.87^n$ . Against this, our explicit constructions (Section 8) are weaker than the exhaustive values already at  $n = 8$ , so they give no useful asymptotic lower bound, and we have *no* construction we can prove beats  $2^{n/2}$ . On the upper side we have only Sperner. We therefore regard Conjecture 9.1 as suggestive rather than well supported, and we are explicit that we prove neither direction.

**Open problem 9.2.** Determine  $f(10)$ . The database has 14 309 547 order types of 10 points ( $\approx 572$  MB), so this is a straightforward but expensive computation, which we did not run.

**Open problem 9.3.** Characterise the maximisers. At  $n = 8$  the maximisers have all maximal subsets of size  $n/2$  and all point-degrees equal; we do not know whether this balanced structure is forced for large  $n$  or is merely typical of small instances.

**Open problem 9.4.** Determine the largest possible number of maximal convex subsets of size exactly 3. Such a subset is a triangle  $T$  with  $P \setminus T \subseteq \mathcal{I}(\text{conv}(T))$ , a strong condition that our data suggests makes them rare (6 at  $n = 5$ , 2 at  $n = 6$ , 1 at  $n = 7$ ).

**Limitations.** (i) Our independent order-type enumerator is incomplete at  $n = 8$  (3313/3315 types); the exact values rely on the external database, which we validated but did not produce. (ii) The upper bound is Sperner’s and is far from sharp. (iii) The constructions in Table 4 are unoptimised; we used simulated annealing on coordinates only as a screening tool and do not claim its outputs are optimal. (iv) No asymptotics are proved.

**Honest score.** We regard the exact values for  $n \leq 9$ , the duality theorem and the reproducible computational pipeline as the strongest parts of this work. The asymptotic question is open, the upper bound is weak, and the paper is a modest but honest computational-structural contribution rather than a breakthrough.

## References

- [1] I. Bárány and P. Valtr. Planar point sets with a small number of empty convex polygons. *Studia Sci. Math. Hungar.* 41(2):243–266, 2004.
- [2] S. W. Bae. Faster counting empty convex polygons in a planar point set. *Inf. Process. Lett.* 175:106221, 2022.
- [3] Erd  
Hos Problem 838 (T. Bloom, ed.). Minimum number of convex subsets determined by  $n$  points. <https://www.erdosproblems.com/838>.
- [4] A. F. Holmsen, H. Nassajian Mojarrad, J. Pach, and G. Tardos. Two extensions of the Erd Hos–Szekeres problem. *J. Eur. Math. Soc.* 22:3741–3763, 2020.
- [5] A. Bukh, J. Matoušek, and G. Nivasch. Lower bounds for weak  $\varepsilon$ -nets and stair-convexity. *Israel J. Math.* 182(1):199–228, 2011. (Conference version: SoCG 2009, 1–10.)
- [6] C. Huemer, D. Oliveros, P. Pérez-Lantero, F. Torra, and B. Vogtenhuber. On weighted sums of numbers of convex polygons in point sets. *Discrete Comput. Geom.* 68, 2022.
- [7] J. S. B. Mitchell, G. Rote, G. Sundaram, and G. Woeginger. Counting convex polygons in planar point sets. *Inf. Process. Lett.* 56(1):45–49, 1995.
- [8] R. Pinchasi, M. Radićić, and M. Sharir. On empty convex polygons in a planar point set. *J. Combin. Theory Ser. A* 113(3):385–419, 2006.
- [9] G. Rote, G. Woeginger, B. Zhu, and Z. Wang. Counting  $k$ -subsets and convex  $k$ -gons in the plane. *Inf. Process. Lett.* 38(3):149–151, 1991.
- [10] G. Rote and G. Woeginger. Counting convex  $k$ -gons in planar point sets. *Inf. Process. Lett.* 41(3):191–194, 1992.
- [11] N. Savova. Point sets with few convex subsets: iterated blow-ups and exact small values. Preprint, 2026. <https://github.com/NikolSavova/erdos-838-convex-subsets>. (Not peer reviewed; cited as adjacent, competing work.)
- [12] A. Suk. On the Erd Hos–Szekeres convex polygon problem. *J. Amer. Math. Soc.* 30(4):1047–1053, 2017.

- [13] O. Aichholzer, F. Aurenhammer, and H. Krasser. Enumerating order types for small point sets with applications. *Order* 19(3):265–281, 2002.
- [14] O. Aichholzer and H. Krasser. The point set order type data base: a collection of applications and results. *CCCG 2001*, 17–20.
- [15] O. Aichholzer and H. Krasser. Abstract order type extension and new results on the rectilinear crossing number. *Comput. Geom.* 36(1):2–15, 2007.
- [16] Aichholzer order-type database. <http://www.ist.tugraz.at/staff/aichholzer/research/rp/triangulations/ordertypes/> (accessed October 2026).